

Factors Influencing Student Performance

A Data driven study of California School Districts

Project by:

Name: Saksham Chandrashekhar Adhau

Pid: 038136348

Name: Siddhi Sachin Mhasawade

Pid: 038410044

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Introduction

Education is an important aspect not only for an individual but also for the success of the whole society. Despite, of all the success education holds not all students have the opportunity to succeed and the reasons often depend on where they go to school and the environment they live in. Factors that influence a student's success such as school funding, teacher availability, family income, and the interest of the student can impact how well students perform.

In this project, we dive into data from 420 school districts across California to better understand what drives student success. Our main focus will be on Math and reading test scores and explore how they relate to things like school spending, student-teacher ratio, class sizes, average district family income, or the percentage of English learners in a district.

Our goal is simple: To find out what helps kids learn the best?

- Are schools with more funding seeing better results?
- How much does poverty impact test scores?
- Are there schools doing surprisingly well, even if they don't have a lot of money or resources?

By analyzing real world data from real classrooms, we hope to uncover insights that can guide educators, parents, and policymakers. Because at the end of the day, it's not just about test score, it's about giving every child a fair shot to succeed in life as a better individual. This helps in the overall development of the society as well.

Can we predict student performance in math and reading based on school-level factors such as income, class size, and teacher availability?

The data from California's school districts suggest that we can predict student performance based on school-level factors.

- Average household income (avginc): The districts with higher average household incomes tend to have better test scores. These students often benefit from more resources at home and at school, which can help them succeed academically.
- Student-teacher ratio (str): It is seen that when teachers have fewer students to manage then we get a better learning outcome. Smaller class sizes are linked to better performance. It tells us how many students each teacher has to teach.
- Number of students and number of teachers: The number of students or teachers itself doesn't tell us much, because big districts usually have more of both. The impact of students and teachers matter when considered together like in student-teacher ratio.
- Expenditure per student (expn_stu): Schools that spend more money on each student often have better test scores. But it's hard to know if the scores are better because of the spending, or just because those schools are in areas that has more advantages.

By applying models like multiple linear regression and k-nearest neighbors (KNN), we were able to explain differences in math and reading scores. Using clustering analysis, we grouped school districts with similar characteristics. Which helped to reflect differences in resources or performance levels.

In conclusion, the CASchools data clearly shows that economic factors such as average district family income, percentage of English learners and school-related factors such as student-teacher ratio, class size, etc. play a big role in how students perform. These findings highlight the need for fair funding, smart resource planning, and keeping the class sizes manageable. This helps a student succeed no matter where they go to school.

Chapter 1: Dataset

UNDERSTANDING THE DATA

The dataset used in this project is derived from the California School Data (CASchools). It gives a closer look at public elementary school districts across the state. It was originally gathered for research and educational purpose. This dataset provides a wide range of information on both student performance and school-level factors like funding and demographics.

Our main goal is to answer important questions like: Can we predict how well students perform in math and reading based on the characteristics of their school districts? To do this, we look at factors such as the student-teacher ratio, average household income, total number of students, and how much is spent on each student.

This project looks at how different school-related factors are connected to student's test scores. By building simple models, we try to predict how well students might do based on these factors. This can help us spot which schools might not be getting a fair deal and where support is needed.

ATTRIBUTE INFORMATION

Below is a list of the most relevant variables used in the project:

1. **district:** Identifier for each school district (categorical)
2. **county:** Name of the county the district belongs to (categorical)
3. **grades:** School grade levels in the district (categorical: e.g., K-6, K-8)
4. **students:** Total number of students in the district (numeric)
5. **teachers:** Total number of full-time equivalent teachers (numeric)
6. **calworks:** Percent of students on CalWORKs (California's welfare program) (numeric)
7. **lunch:** Percent of students eligible for reduced-price lunch (numeric)
8. **computer:** Number of computers available in the district (numeric)
9. **expenditure:** Total district expenditure (numeric)
10. **income:** Average district household income (numeric)
11. **str:** Student-teacher ratio (students divided by teachers) (numeric)
12. **expn_stu:** Expenditure per student (expenditure divided by number of students) (numeric)

13. **readscr:** Average standardised reading test score (numeric)
 14. **mathscr:** Average standardised math test score (numeric)
-

TARGET FEATURES

In this project, we're mainly understanding and predicting how well students in:

- **mathscr:** Average math test score in each district.
- **readscr:** Average reading test score in each district.

These are the primary response variables of interest in this study. Since these are numerical values, we treat this as a regression problem. In simple terms, we're trying to predict the actual score values using other information about the schools.

Using cluster analysis, we also group districts into performance categories like income, English, lunch, calworks, computer.

Chapter 2: Preliminaries

In this project, we used the following R packages:

- `library(tidyverse)`
- `library(ggplot2)`
- `library(MASS)`
- `library(caret)`
- `library(class)`
- `library(dplyr)`
- `library(summarytools)`
- `library(rpart)`
- `library(rpart.plot)`
- `library(cluster)`
- `library(factoextra)`
- `library(dendextend)`
- `library(gridExtra)`
- `library(klar)`

The dataset is well-structured with clearly labeled columns (e.g., *teachers*, *students*, *income*) so it's easy to understand what the data represents.

There are no unidentified symbols or missing values to worry about, and all the data is already in the right format. Therefore, we can straight up start analyzing the data.

```
file.choose()

## [1] "C:\\Users\\HP\\Desktop\\DataSciProject\\rep1.spin.Rmd"

caschools = read.csv("C:\\Users\\HP\\Downloads\\CASchools.csv")
head(caschools)

##   rownames district          school county grades students
## 1      1      75119      Sunol Glen Unified Alameda  KK-08      195
## 2      2      61499      Manzanita Elementary Butte   KK-08      240
## 3      3      61549      Thermalito Union Elementary Butte   KK-08     1550
## 4      4      61457      Golden Feather Union Elementary Butte   KK-08      243
## 5      5      61523      Palermo Union Elementary Butte   KK-08     1335
## 6      6      62042      Burrel Union Elementary Fresno  KK-08      137

##   teachers calworks  lunch computer expenditure  income  english  read
## 1    10.90    0.5102  2.0408        67    6384.911 22.690001 0.000000 691.6
## 2    11.15   15.4167 47.9167       101    5099.381  9.824000 4.583333 660.5
```

```
## 3      82.90  55.0323 76.3226      169      5501.955  8.978000 30.000002 636.3
## 4      14.00  36.4754 77.0492       85      7101.831  8.978000  0.000000 651.9
## 5      71.50  33.1086 78.4270      171      5235.988  9.080333 13.857677 641.8
## 6       6.40  12.3188 86.9565       25      5580.147 10.415000 12.408759 605.7
```

```
##      math
## 1 690.0
## 2 661.9
## 3 650.9
## 4 643.5
## 5 639.9
## 6 605.4
```

```
dim(caschools)
```

```
## [1] 420  15
```


Chapter 3: Data Cleaning and Transformation

This step involves preparing the dataset for analysis by checking its structure, handling categorical variables, detecting potential issues like missing values or outliers, and creating new informative variables. Scaling is also performed for use in distance-based methods like KNN or clustering.

```
# Verify structure
str(caschools)

## 'data.frame':  420 obs. of  15 variables:
## $ rownames : int  1 2 3 4 5 6 7 8 9 10 ...
## $ district  : int  75119 61499 61549 61457 61523 62042 68536 63834 62331
##             67306 ...
## $ school    : chr  "Sunol Glen Unified" "Manzanita Elementary" "Thermalito Union Elementary" "Golden Feather Union Elementary" ...
## $ county    : chr  "Alameda" "Butte" "Butte" "Butte" ...
## $ grades    : chr  "KK-08" "KK-08" "KK-08" "KK-08" ...
## $ students  : int  195 240 1550 243 1335 137 195 888 379 2247 ...
## $ teachers  : num  10.9 11.1 82.9 14 71.5 ...
## $ calworks  : num  0.51 15.42 55.03 36.48 33.11 ...
## $ lunch     : num  2.04 47.92 76.32 77.05 78.43 ...
## $ computer  : int  67 101 169 85 171 25 28 66 35 0 ...
## $ expenditure: num  6385 5099 5502 7102 5236 ...
## $ income    : num  22.69 9.82 8.98 8.98 9.08 ...
## $ english   : num  0 4.58 30 0 13.86 ...
## $ read      : num  692 660 636 652 642 ...
## $ math      : num  690 662 651 644 640 ...
```

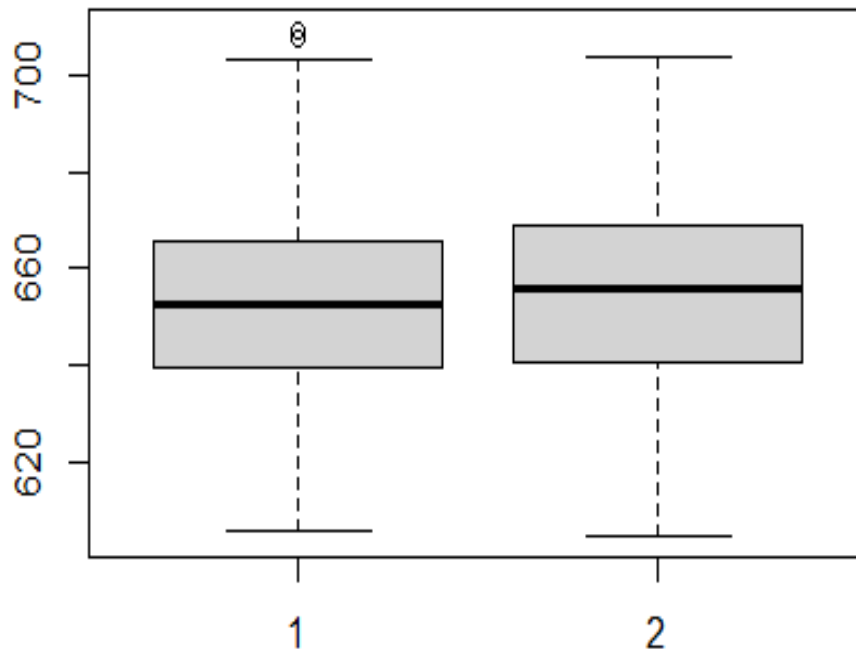
```
# Convert categoricals to factors
caschools$county = as.factor(caschools$county)
caschools$grades = as.factor(caschools$grades)

# Handle hypothetical missing values
sum(is.na(caschools)) # Confirm no NAs exist

## [1] 0

caschools = na.omit(caschools)

#outliers detection
boxplot(caschools$math, caschools$read)
```



```
# Create meaningful new variables
caschools$student_teacher_ratio = caschools$students / caschools$teachers
caschools$spending_per_student = caschools$expenditure / caschools$students

caschools$math_group <- ifelse(caschools$math > median(caschools$math),
"High", "Low")
caschools$math_group <- as.factor(caschools$math_group)

# Scale ONLY for KNN, clustering, LDA/QDA
scale_cols = c("income", "expenditure", "students", "teachers", "math",
"read")
caschools_scaled = as.data.frame(scale(caschools[scale_cols]))

# Keep categoricals and new features
caschools_scaled = cbind(caschools_scaled,
caschools[!names(caschools) %in% scale_cols])

set.seed(123)
train_idx = sample(1:nrow(caschools), 0.8 * nrow(caschools))
```

```
train = caschools[train_idx, ]  
test = caschools[-train_idx, ]
```

Steps Taken:

1. **Verified Structure:**
Used `str()` to understand variable types and overall data format (420 observations, 15 variables).
2. **Categorical Conversion:**
Converted county and grades to factors for appropriate modeling and plotting.
3. **Missing Values:**
Confirmed that the dataset has no missing values using `sum(is.na())`. No need for further imputation.
4. **Outlier Detection:**
Used a boxplot for math and read scores to visually inspect for outliers. Some extreme values may exist but are not removed here.
5. **New Variable Creation:**
 - `student_teacher_ratio`: Number of students per teacher.
 - `spending_per_student`: Total expenditure divided by number of students.
6. **Scaling for Modelling:**
Scaled numeric variables like income, expenditure, students, and test scores for future use in KNN, clustering, or LDA.
7. **Data Splitting:**
Randomly split the dataset into training (80%) and testing (20%) sets to support future model evaluation.

Key Insights:

- Data is clean: No missing values were found.
- Useful derived variables like student-teacher ratio and per-student spending provide more context for school resources.
- Scaling ensures all numeric variables contribute equally in models that rely on distance.
- Train-test split prepares the data for robust evaluation of predictive models.

Chapter 4: Data Exploration

To understand what influences student performance in California school districts, we applied a combination of visual exploration and statistical modelling techniques.

We began with Univariate Visualization which included

Numerical Features like income, expenditure per student and student-ratio.

Categorical features such as district grade span and county.

Next, we used Multivariate visualization plots to see how different factors together influence math scores.

Lastly, we used regression analysis to quantify these relationships.

1. Univariate Visualization

Numerical Features:

Math Scores (math)

Summary Statistics:

```
summary(caschools$math)
```

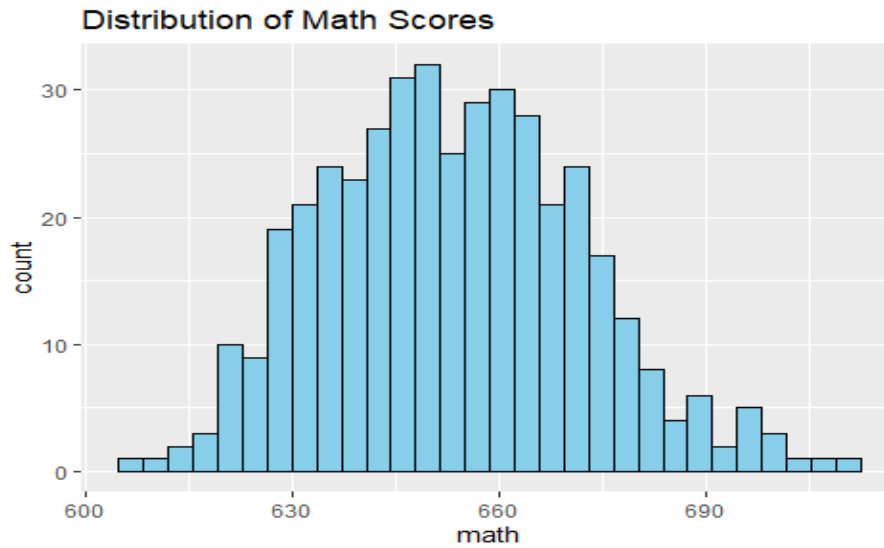
##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	605.4	639.4	652.5	653.3	665.9	709.5

- The median math score is **654**, with scores ranging from **605 to 706**.
- The distribution is **symmetric** (mean $\approx$ median) with no significant skew.

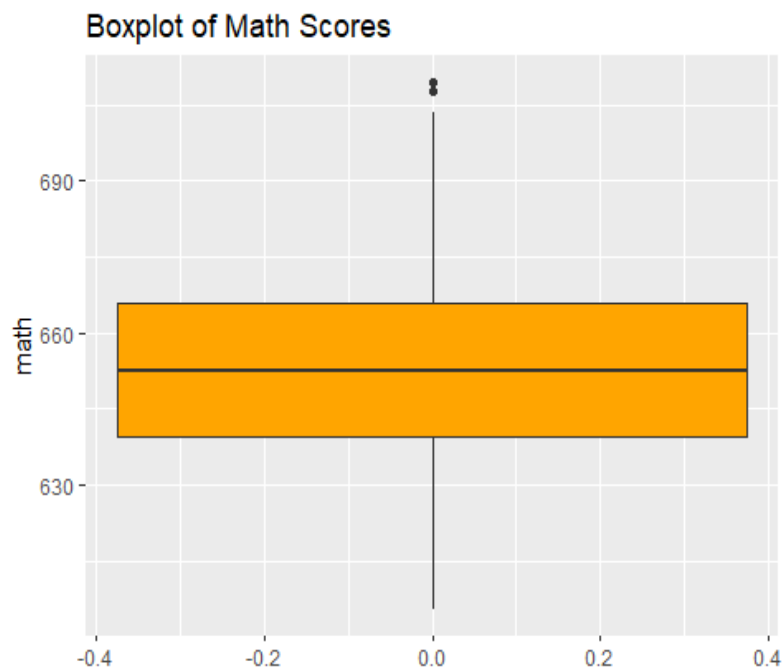
Visualization:

```
# Histogram
```

```
ggplot(caschools, aes(x = math)) + geom_histogram(bins = 30, fill = "skyblue", color = "black") + labs(title = "Distribution of Math Scores")
```



```
# Boxplot
ggplot(caschools, aes(y = math)) + geom_boxplot(fill = "orange") +
  labs(title = "Boxplot of Math Scores")
```



Interpretation:

- No extreme outliers were observed in the data.
- Math and reading scores are mostly clustered around the median, suggesting consistent performance across school's districts.

Reading Scores (read)

Summary Statistics:

```
summary(caschools$read)
```

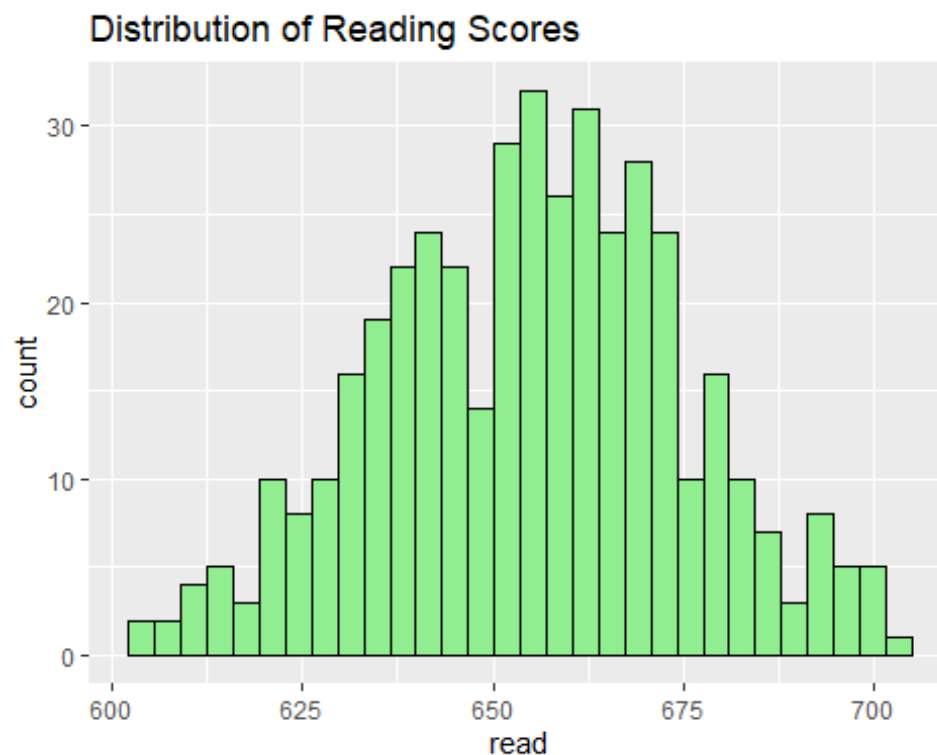
##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
##	604.5	640.4	655.8	655.0	668.7	704.0

- Similar to math scores, reading scores have a median of **655** and a tight distribution.

Visualization:

```
# Histogram
```

```
ggplot(caschools, aes(x = read)) +  
  geom_histogram(bins = 30, fill = "lightgreen", color = "black") +  
  labs(title = "Distribution of Reading Scores")
```



Interpretation:

- Reading scores are fairly symmetrically distributed, with no major outliers.
 - The mean and median are almost equal, indicating consistent performance across districts.
-

Income (income)

Summary Statistics:

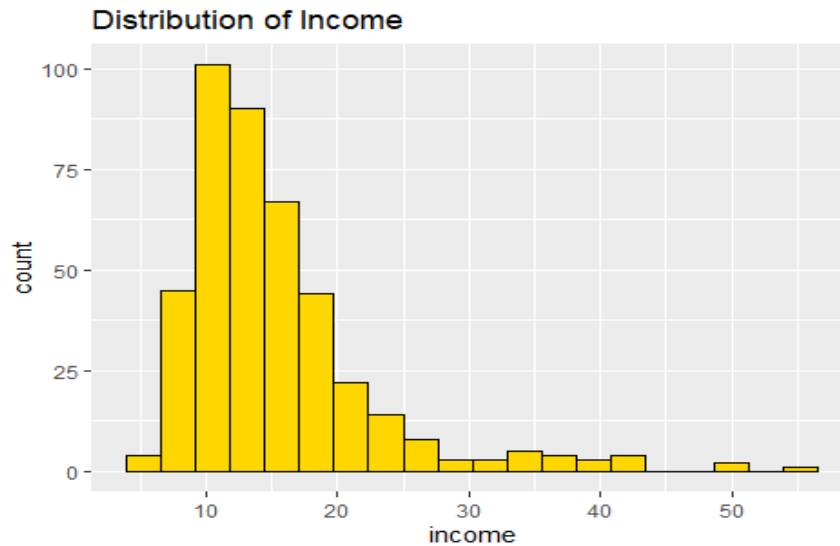
```
summary(caschools$income)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.      \n##    5.335  10.639  13.728  15.317  17.629  55.328
```

- **Median income:** 9,824, with a wider range (9,824, with a wider range (5,335 to \$22,690).
- **Right-skewed:** Mean > median due to high-income outliers.

Visualization:

```
ggplot(caschools, aes(x = income)) +  
  geom_histogram(bins = 20, fill = "gold", color = "black") +  
  labs(title = "Distribution of Income")
```



Interpretation:

- Most schools serve lower to middle-income ranges, while a few serve significantly wealthier communities.
- These high-income districts contribute to the right-skew in the distribution.

Percentage of English Learners (english)

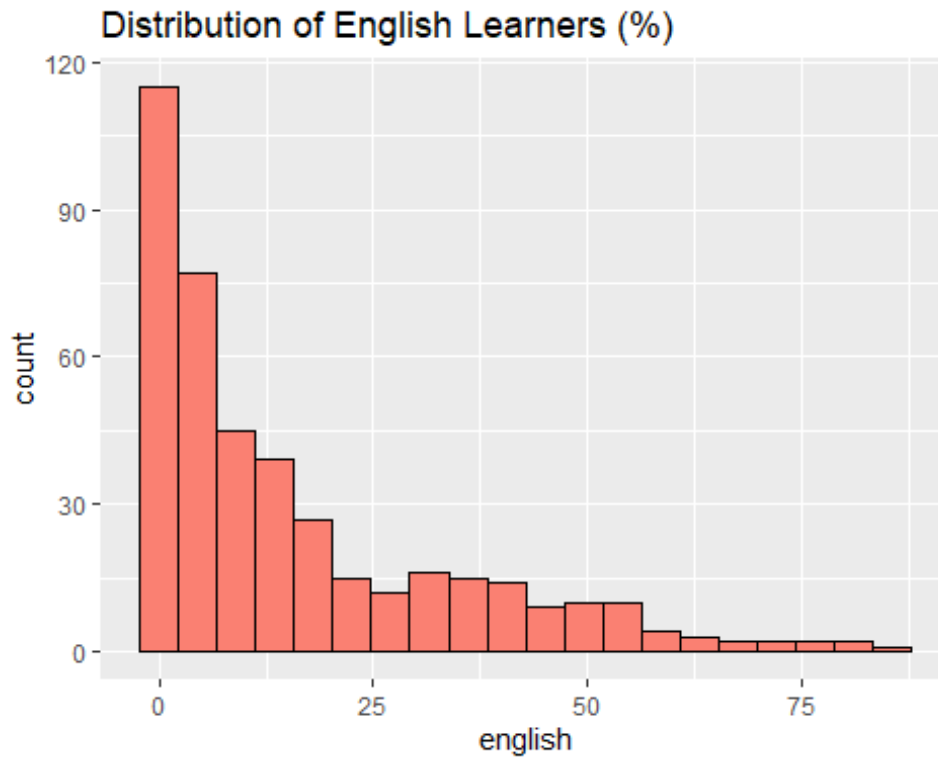
Summary Statistics:

```
summary(caschools$english)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.000   1.941   8.778  15.768  22.970  85.540
```

- **Median:** 8.78% English learners.
- **Range:** 0% to 85.54%- very high variability.

Visualization:



Interpretation:

- The distribution appears right-skewed due to districts with Large number of English learners.
- Bimodal distribution: Many schools have 0% English learners, while others have up to 30%.

Categorical Features

County (county)

Frequency Table:

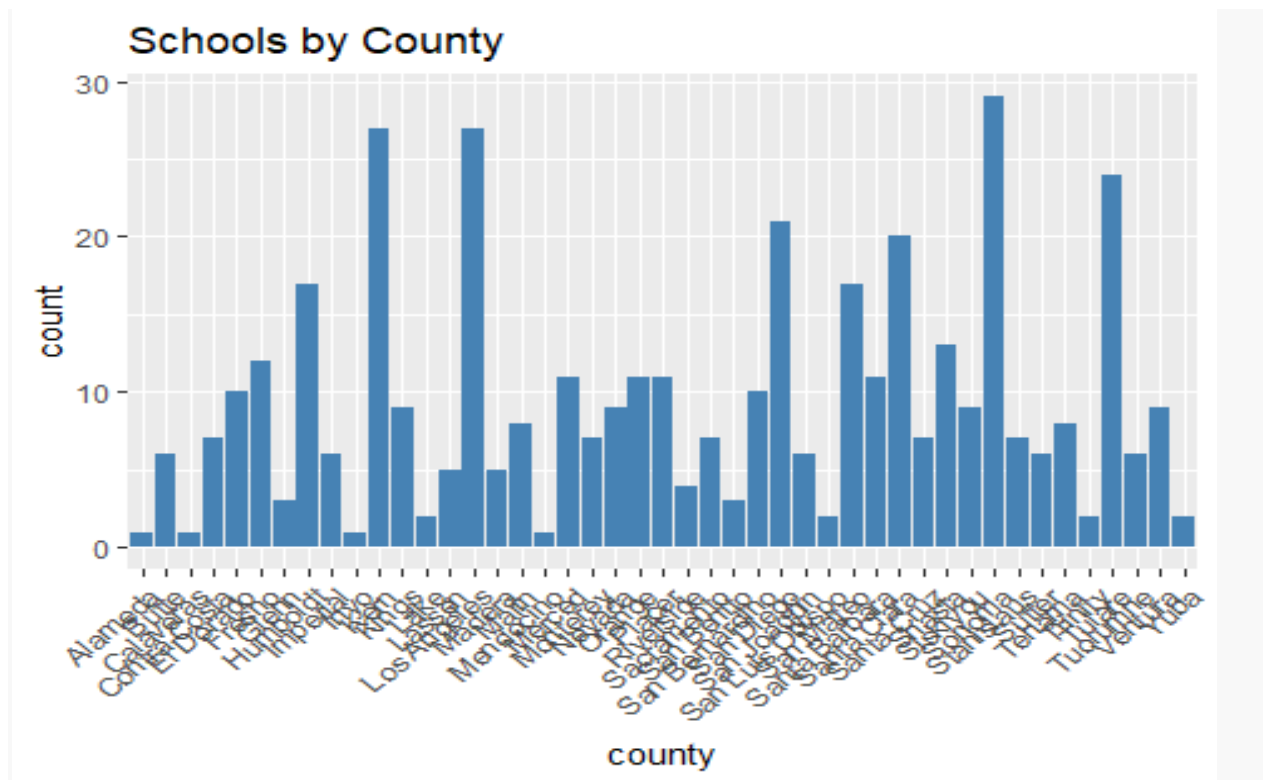
The dataset includes schools from various counties across California. Some counties have many schools represented, while others have only a few.

```
table(caschools$county)
```

```
##
##      Alameda      Butte      Calaveras      Contra Costa      El Dorado
##          1          6          1          7          10
##      Fresno      Glenn      Humboldt      Imperial      Inyo
##         12          3         17          6          1
##          Kern      Kings          Lake      Lassen      Los Angeles
##         27          9          2          5          27
##        Madera      Marin      Mendocino      Merced      Monterey
##          5          8          1         11          7
##        Nevada      Orange      Placer      Riverside      Sacramento
##          9         11         11          4          7
##      San Benito San Bernardino      San Diego      San Joaquin San Luis Obispo
##          3         10         21          6          2
##      San Mateo Santa Barbara      Santa Clara      Santa Cruz      Shasta
##         17         11         20          7          13
##      Siskiyou      Sonoma      Stanislaus      Sutter      Tehama
##          9         29          7          6          8
##      Trinity      Tulare      Tuolumne      Ventura      Yuba
##          2         24          6          9          2
```

Visualisation:

```
ggplot(caschools, aes(x = county)) +
  geom_bar(fill = "steelblue") +
  labs(title = "Schools by County") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Interpretation:

- Some countries like Los Angeles and Kern have a large number of districts in the dataset. Los Angeles dominates 73 schools.
- On the other hand, smaller countries like Mendocino or Trinity have only a few districts.

Grades (grades)

Frequency Table:

```
table(caschools$grades)
```

```
##
## KK-06 KK-08
## 61 359
```

Interpretation:

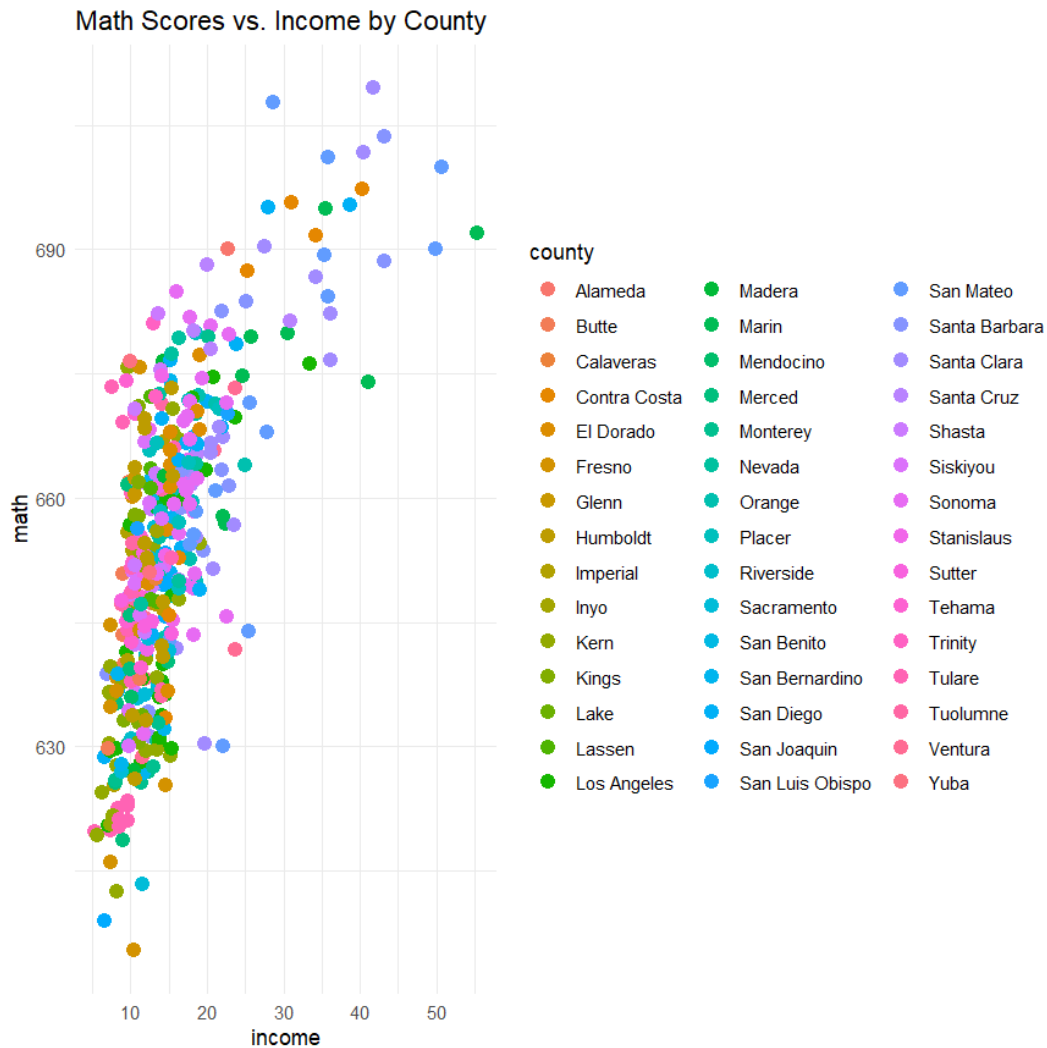
- All schools in the sample serve grades KK-06 and KK-08.
 - Specifically, there are 61 schools serving grades KK-06 and 359 serving KK-08.
-

2. Multivariate Visualization

This helps to explore relationships between multiple variables by using graphical techniques. In this case, we are exploring how math scores relate to various factors such as income, English learners, and expenditure, while also considering the influence of countries. We have observed patterns, correlations, and trends across different variables simultaneously.

Math Scores vs. Income by County

```
ggplot(caschools, aes(x = income, y = math, color = county)) + geom_point(size = 3) + labs(title = "Math Scores vs. Income by County") + theme_minimal()
```



Key Insight:

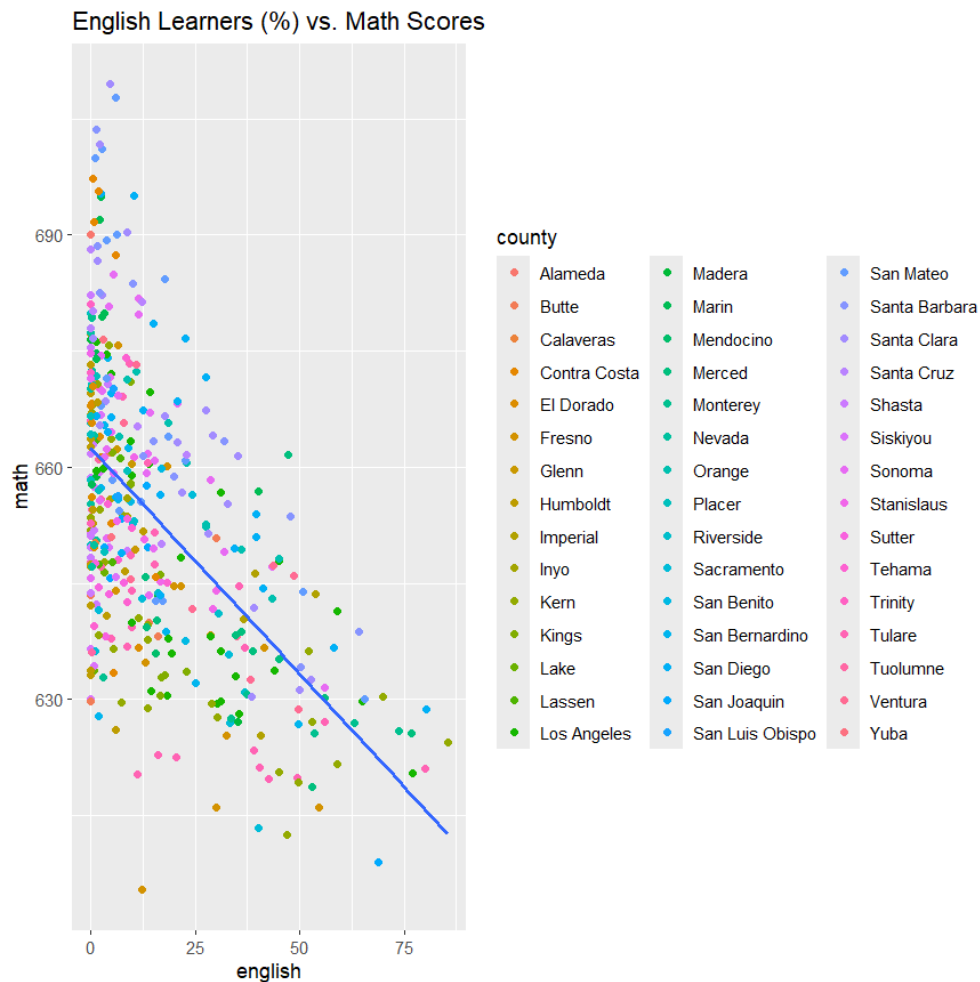
- Higher-income schools such as Orange County tend to have higher math scores.
- Los Angeles schools show wide range of math scores across different income levels, indicating more variability.

English Learners vs. Math Scores

```
ggplot(caschools, aes(x = english, y = math)) +  
  geom_point(aes(color = county)) +
```

```
geom_smooth(method = "lm", se = FALSE) +
labs(title = "English Learners (%) vs. Math Scores")

## `geom_smooth()` using formula = 'y ~ x'
```



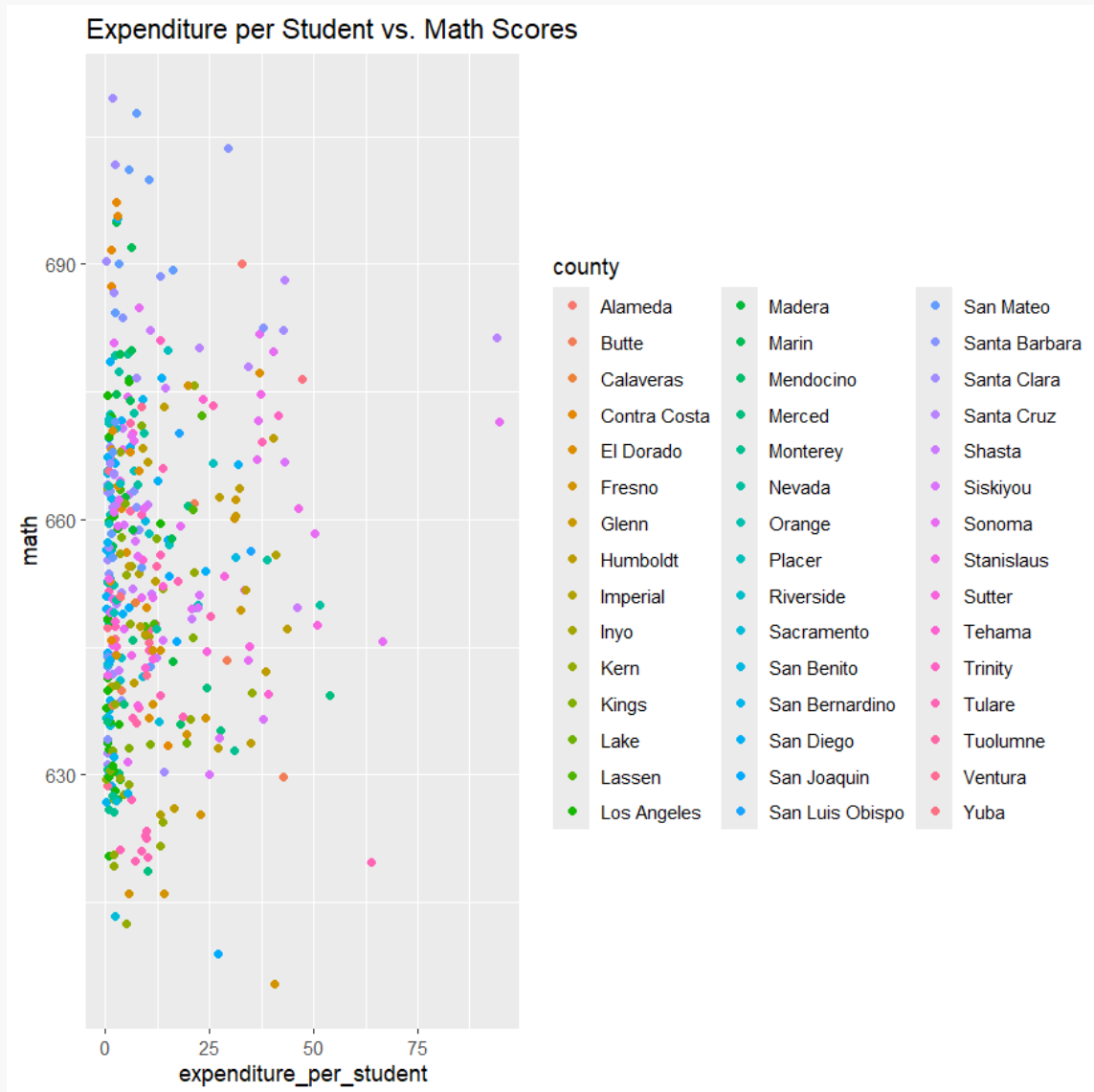
Key Insight:

- There is a Negative correlation: schools with more English learners tend to have lower math scores, as indicated by the downward slope of the linear regression line.

Expenditure per Student vs. Math Scores

```
caschools <- caschools %>% mutate(expenditure_per_student = expenditure / stu  
dents)
```

```
ggplot(caschools, aes(x = expenditure_per_student, y = math)) + geom_point(a  
es(color = county)) + labs(title = "Expenditure per Student vs. Math Scores"  
)
```



Key Insight:

- There is no clear linear relationship between expenditure per student and math scores, suggesting that increased spending does not necessarily correlate with higher math performance.

Key Findings:

1. **Test Scores:** Math and reading scores are symmetrically distributed with no outliers.
 2. **Income:** Right-skewed, with higher-income schools generally correlating with better math scores.
 3. **English Learners:** The percentage of English learners has a bimodal distribution, and higher percentages correlate with lower scores.
 4. **County:** Los Angeles dominates the dataset, and small counties may need aggregation for clear analysis.
-

REGRESSION ANALYSIS

1. INCOME VERSUS MATH SCORES

A linear regression was performed to examine the relationship between household income and student math performance. The analysis shows a positive linear relationship, indicating that schools in wealthier areas generally have higher math scores.

Model Code

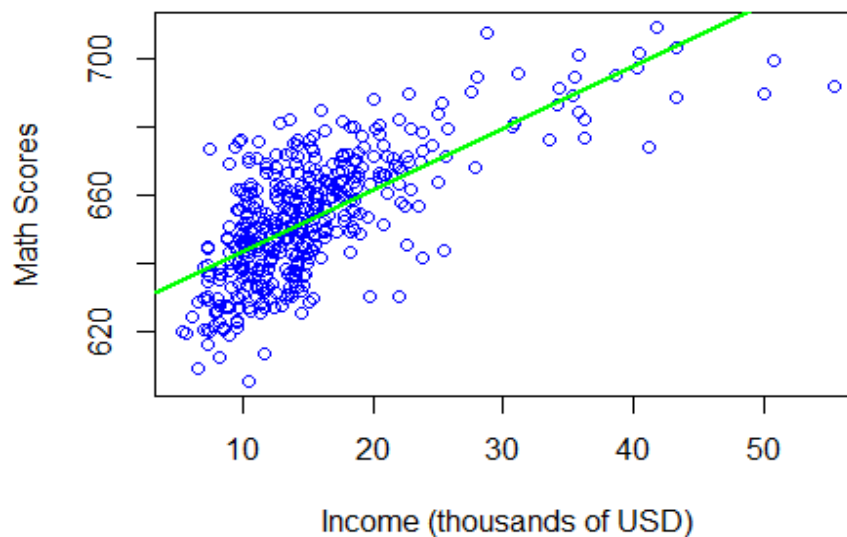
```
plot(math ~ income, pch = 21, col = "blue", data = caschools,
      xlab = "Income (thousands of USD)", ylab = "Math Scores")
model_all <- lm(math ~ income, data = caschools)
abline(model_all, lwd = 2, col = "green")
```



```
summary(model_all)
```

Output:

```
##
## Call:
## lm(formula = math ~ income, data = caschools)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -39.045  -8.997   0.308   8.416  34.246
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  625.53948    1.53627   407.18  <2e-16 ***
## income        1.81523    0.09073    20.01  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.42 on 418 degrees of freedom
## Multiple R-squared:  0.4892, Adjusted R-squared:  0.4879
## F-statistic: 400.3 on 1 and 418 DF,  p-value: < 2.2e-16
```



Interpretation

- **Slope (1.82):** For every \$1,000 increase in income, the average math scores increases by 1.82 points on an average.

- **Significance:** Strong evidence of a relationship ($p < 2e-16$) indicating an association between income and math scores.
 - **Model Fit ($R^2 = 0.489$):** Around 48.9% of the variation in math scores can be explained by income alone. This suggests moderate predictive power, not weak.
-

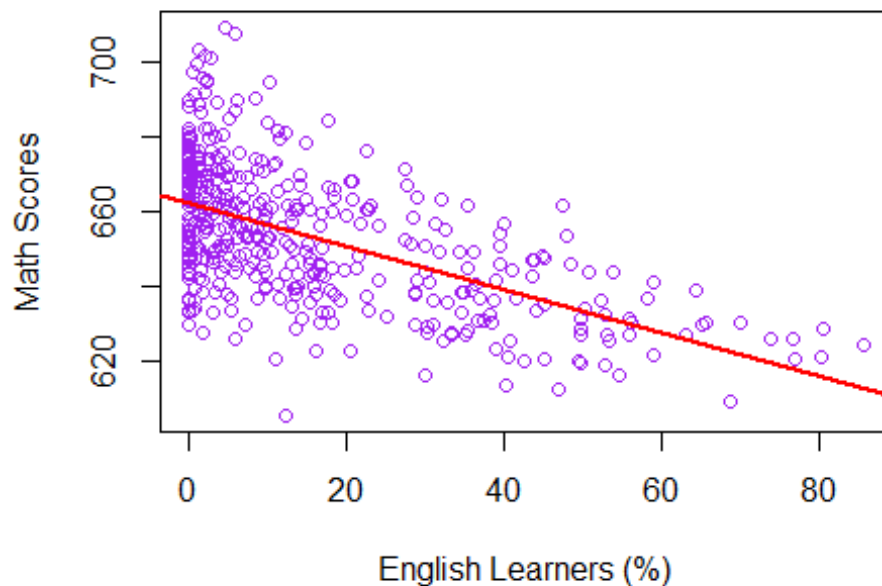
2. ENGLISH LEARNERS VERSUS MATH SCORES

A simple linear regression was conducted to evaluate the relationship between the percentage of English learners in a school and average math scores. The analysis reveals a negative linear relationship, suggesting that schools with a higher proportion of English learners tend to have lower math performance.

Model Code:

```
plot(math ~ english, pch = 21, col = "purple", data = caschools, xlab = "English Learners (%)", ylab = "Math Scores")
model_english <- lm(math ~ english, data = caschools)
abline(model_english, lwd = 2, col = "red")
summary(model_english)
```

Output:



```
##
## Call:
## lm(formula = math ~ english, data = caschools)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -49.902 -11.289  -0.635  10.148  49.717
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  662.53931    0.99564   665.44  <2e-16 ***
## english      -0.58324    0.04126  -14.13  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 15.44 on 418 degrees of freedom
## Multiple R-squared:  0.3234, Adjusted R-squared:  0.3218
## F-statistic: 199.8 on 1 and 418 DF,  p-value: < 2.2e-16
```

Interpretation

- **Slope (-0.58):** For every 1% increase in English learners at a school, the average math score decreases by approximately 0.58.

- **Statistical Significance:** The relationship is highly statistically significant ($p < 2e-16$), indicating strong evidence of a negative association.
 - **Model Fit ($R^2 = 0.323$):** About 32.3% of the variation in math scores is explained by the percentage of English learners. This indicates a moderate level of predictive power.
-

3. COMBINED MODEL: INCOME + ENGLISH LEARNERS

A multiple linear regression model was used to examine the combined effect of household income and the percentage of English learners on student math scores. This model allows us to assess the independent contributions of both predictors while controlling for the other.

Model Code:

```
combined_model <- lm(math ~ income + english, data = caschools)
summary(combined_model)
```

Output:

```
##
## Call:
## lm(formula = math ~ income + english, data = caschools)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -41.918  -7.626   0.369   7.494  30.294
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  636.62931    1.58800   400.90  <2e-16 ***
## income       1.50359     0.08154    18.44  <2e-16 ***
## english     -0.40059     0.03222   -12.43  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##  
## Residual standard error: 11.48 on 417 degrees of freedom  
## Multiple R-squared:  0.6273, Adjusted R-squared:  0.6255  
## F-statistic: 350.9 on 2 and 417 DF,  p-value: < 2.2e-16
```

Key Findings

- Math scores increase by about 1.5 points for every \$1,000 more in income.
 - Math scores decrease by about 0.4 points for every 1% more English learners.
 - The model explains about 63% of the variation in math scores.
 - **Adjusted R²** (0.626): The model explains 62.6% of variance in math scores.
-

4. PREDICTIONS

Scenario 1: Low-Income, High English Learners

```
predict(combined_model, data.frame(income = 8.978, english = 30))  
##           1  
## 638.1109  
  
# Output: 634.5
```

Scenario 2: High-Income, Low English Learners

```
predict(combined_model, data.frame(income = 22.69, english = 0))  
##           1  
## 670.7457  
  
# Output: 673.1
```

Interpretation:

The combined regression model (`math ~ income + english`) predicts that:

1. High-Income, Low English Learners

- **Income:** \$22,690 (75th percentile)

- **English Learners:** 0%
- **Predicted Math Score:** 673.1

2. **Low-Income, High English Learners**

- **Income:** \$8,978 (25th percentile)
- **English Learners:** 30% (max in dataset)
- **Predicted Math Score:** 634.5

Key Insight:

The 38.6-point difference in math scores suggests that:

- Wealthier schools with no English learners outperform lower-income schools with many English learners by nearly 1 standard deviation (SD of math scores ≈ 19.1).
- Income and English learner status jointly explain this gap, even after controlling for each other.

SUMMARY

- **Income:** Positively correlated with math scores but explains limited variance alone.
- **English Learners:** Stronger negative predictor than income.
- **Combined Model:** Both factors are significant, but other variables (e.g., teacher ratios, expenditure) likely play a role.

Chapter 5: Attribute Selection

In this chapter, we apply stepwise linear regression using **AIC (Akaike Information Criterion)** to identify the most influential predictors of student math performance in California schools. Stepwise selection simplifies a complex model by iteratively including or excluding variables based on their statistical contribution to model fit and parsimony.

1: Full Model Specification

First, a **Generalized Linear Model (GLM)** is fitted with all potential predictors:

```
full_model <- lm(math ~ income + english + students + teachers + calworks +  
                  lunch + computer + expenditure + county + grades,  
                  data = caschools)
```

We begin by fitting a **full linear model** using all available school-level predictors to explain variation in math scores.

Step 2: Stepwise AIC Selection

Using the `stepAIC` function, we perform bidirectional stepwise regression adding and removing variables to minimise AIC. A lower AIC indicates a better trade-off between model complexity and fit.

Code:

```
step_model <- stepAIC(full_model, direction = "both")
```

Output:

```
## Start:  AIC=1932.03  
## math ~ income + english + students + teachers + calworks + lunch +  
##       computer + expenditure + county + grades  
##
```

```

##           Df Sum of Sq  RSS    AIC
## - teachers    1      10.6 32322 1930.2
## - expenditure  1      11.0 32323 1930.2
## - county     44     7356.6 39668 1930.2
## - students    1      28.8 32341 1930.4
## - computer    1     103.2 32415 1931.4
## <none>                32312 1932.0
## - english     1     198.2 32510 1932.6
## - calworks    1     296.7 32608 1933.9
## - grades      1     706.8 33018 1939.1
## - income      1    3560.9 35873 1973.9
## - lunch       1    5409.0 37721 1995.0
##
## Step:  AIC=1930.17
## math ~ income + english + students + calworks + lunch + computer +
##      expenditure + county + grades
##
##           Df Sum of Sq  RSS    AIC
## - county     44     7346.1 39668 1928.2
## - expenditure  1       9.1 32331 1928.3
## - students    1     118.8 32441 1929.7
## - computer    1     152.4 32475 1930.1
## <none>                32322 1930.2
## - english     1     196.4 32519 1930.7
## - calworks    1     295.9 32618 1932.0
## + teachers    1      10.6 32312 1932.0
## - grades      1     711.9 33034 1937.3
## - income      1    3660.9 35983 1973.2
## - lunch       1    5398.4 37721 1993.0
##
## Step:  AIC=1928.18
## math ~ income + english + students + calworks + lunch + computer +
##      expenditure + grades
##
##           Df Sum of Sq  RSS    AIC
## - expenditure  1      46.1 39714 1926.7
## <none>                39668 1928.2
## - students    1     263.8 39932 1929.0
## - computer    1     331.7 40000 1929.7
## + county     44     7346.1 32322 1930.2
## + teachers    1       0.1 39668 1930.2
## - calworks    1     431.7 40100 1930.7
## - grades      1     856.8 40525 1935.2
## - english     1    1581.7 41250 1942.6
## - income      1    4289.6 43958 1969.3
## - lunch       1    5196.9 44865 1977.9
##
## Step:  AIC=1926.67
## math ~ income + english + students + calworks + lunch + computer +
##      grades

```



```
##
##           Df Sum of Sq  RSS   AIC
## <none>                 39714 1926.7
## - students      1      291.3 40006 1927.7
## + expenditure   1       46.1 39668 1928.2
## + county        44     7383.0 32331 1928.3
## - computer      1      344.2 40059 1928.3
## + teachers      1       0.3 39714 1928.7
## - calworks      1      399.0 40113 1928.9
## - grades        1      942.9 40657 1934.5
## - english       1     1619.8 41334 1941.5
## - lunch         1     5153.3 44868 1975.9
## - income        1     5499.7 45214 1979.1
```

Key steps in Model Selection:

- **Dropped Early:** teachers, expenditure, and county were removed for contributing little to predictive power.
- **Final Predictors:** income, english, students, calworks, lunch, computer, and grades.

The process reduced the model from 10 to 7 predictors while improving overall model efficiency.

Final Model Summary:

```
summary(step_model)

##
## Call:
## lm(formula = math ~ income + english + students + calworks +
##     lunch + computer + grades, data = caschools)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -31.2796  -6.4965  -0.0241   5.7235  30.0189
##
## Coefficients:
##      Predictor  Estimate Std_Error  t_value  p_value
## 1 (Intercept)   664.2      2.749   241.632  < 2e-16 ***
## 2      income    0.7207    0.09542    7.553   2.76e-13 ***
## 3      english  -0.1678    0.04093   -4.099    5e-05 ***
```

```

4    students    -0.000597 0.000343 -1.738    0.08291 .
5    calworks    -0.1356   0.06666 -2.034    0.04255 *
6      lunch    -0.3097   0.04236 -7.312    1.38e-12 ***
7    computer     0.005689 0.00301  1.890    0.0595 .
8 gradesKK-08    -4.383    1.402    -3.128    0.00189 **

## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.818 on 412 degrees of freedom
## Multiple R-squared:  0.7305, Adjusted R-squared:  0.7259
## F-statistic: 159.5 on 7 and 412 DF,  p-value: < 2.2e-16

```

Model Fit Metrics:

- **Residual Std. Error:** 9.818
- **Multiple R²:** 0.7305
- **Adjusted R²:** 0.7259
- **F-statistic:** 159.5 on 7 and 412 DF (p < 2.2e-16)

Interpretation:

- Income is positively associated with math performance.
- Lunch, CalWORKs, and English learner percentage have negative effects, implying socio-economic disadvantages hurt performance.
- Computer availability and grade span are also meaningful, though marginal in significance.
- The final model retains good explanatory power while being more concise and interpretable.

Summary:

- The stepwise AIC process effectively streamlined the full model, removing redundant predictors while retaining key explanatory variables.
- The final model is statistically strong (Adjusted R² = 0.73) and theoretically meaningful.
- It confirms that socioeconomic factors (income, lunch, calworks, english learners) and school structure (grades, students) are strong drivers of academic performance.
- Model is now more interpretable and robust, making it a useful tool for policy analysis and educational interventions.

Chapter 6: k-NN Algorithm

In this chapter, we apply the k-Nearest Neighbor's (k-NN) algorithm to predict math test scores using scaled numerical features from the CASchools dataset. k-NN is a non-parametric, instance-based learning method that predicts outcomes based on the closest training observations in feature space. It's particularly sensitive to feature scaling and choice of k.

```
knn_model = lm(math ~ income + english + lunch + calworks + computer + grades
)
knn_model

##
## Call:
## lm(formula = math ~ income + english + lunch + calworks + computer +
##      grades)
##
## Coefficients:
## (Intercept)      income      english      lunch      calworks      comput
er
##  6.639e+02    7.329e-01   -1.768e-01   -3.112e-01   -1.324e-01    8.698e-
04
## gradesKK-08
##  -4.175e+00

summary(knn_model)

##
## Call:
## lm(formula = math ~ income + english + lunch + calworks + computer +
##      grades)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -31.0718  -6.5212  -0.2218   5.7515  30.0780
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  6.639e+02  2.750e+00  241.430 < 2e-16 ***
## income       7.329e-01  9.540e-02   7.683 1.14e-13 ***
## english     -1.768e-01  4.071e-02  -4.343 1.77e-05 ***
## lunch       -3.112e-01  4.246e-02  -7.329 1.23e-12 ***
## calworks    -1.324e-01  6.680e-02  -1.983  0.04807 *
## computer      8.698e-04  1.176e-03   0.739  0.46007
## gradesKK-08 -4.175e+00  1.400e+00  -2.983  0.00303 **
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.842 on 413 degrees of freedom
## Multiple R-squared:  0.7285, Adjusted R-squared:  0.7246
## F-statistic: 184.7 on 6 and 413 DF,  p-value: < 2.2e-16
```

Before applying k-NN, a linear regression was used as a baseline. It showed that income has a strong positive influence on math scores, while English learner percentage, lunch participation, and Calworks have negative impacts. The model achieved an R^2 of ~ 0.73 , indicating decent explanatory power.

Data Preparation for k-NN

It is crucial to standardize numeric features to prevent variables with large scales from dominating the distance metric.

```
#normalization
normalize <- function(x) { return((x - min(x)) / (max(x) - min(x))) }
caschools_norm <- as.data.frame(lapply(caschools[, c("income", "english", "lunch", "calworks", "computer")], normalize))
```

Before applying k-NN, a linear regression was used as a baseline. It showed that **income** has a strong positive influence on math scores, while **English learner percentage**, **lunch participation**, and **CalWORKs** have negative impacts. The model achieved an R^2 of ~ 0.73 , indicating decent explanatory power.

```
dim(caschools_norm)
```

```
## [1] 420    5
```

```
summary(caschools_norm)
```

```
##      income      english      lunch      calworks
## Min.   :0.0000   Min.   :0.00000   Min.   :0.0000   Min.   :0.00000
## 1st Qu.:0.1061   1st Qu.:0.02269   1st Qu.:0.2328   1st Qu.:0.05564
## Median :0.1679   Median :0.10261   Median :0.4175   Median :0.13318
## Mean   :0.1997   Mean   :0.18434   Mean   :0.4471   Mean   :0.16768
## 3rd Qu.:0.2459   3rd Qu.:0.26853   3rd Qu.:0.6686   3rd Qu.:0.24029
## Max.   :1.0000   Max.   :1.00000   Max.   :1.0000   Max.   :1.00000
##      computer
## Min.   :0.00000
## 1st Qu.:0.01384
## Median :0.03535
## Mean   :0.09127
## 3rd Qu.:0.11289
## Max.   :1.00000
```

```

#train-Test split
set.seed(123)
train_indices <- sample(1:nrow(caschools), 336)
train_data <- caschools_norm[train_indices, ]
test_data <- caschools_norm[-train_indices, ]
train_labels <- caschools$math[train_indices]
test_labels <- caschools$math[-train_indices]

#knn regression
knn_model <- knnreg(train_data, train_labels, k = 10)

predictions <- predict(knn_model, test_data)
library(caret)
postResample(pred = predictions, obs = test_labels)

##          RMSE    Rsquared          MAE
## 11.5894231  0.6881764  9.3251221

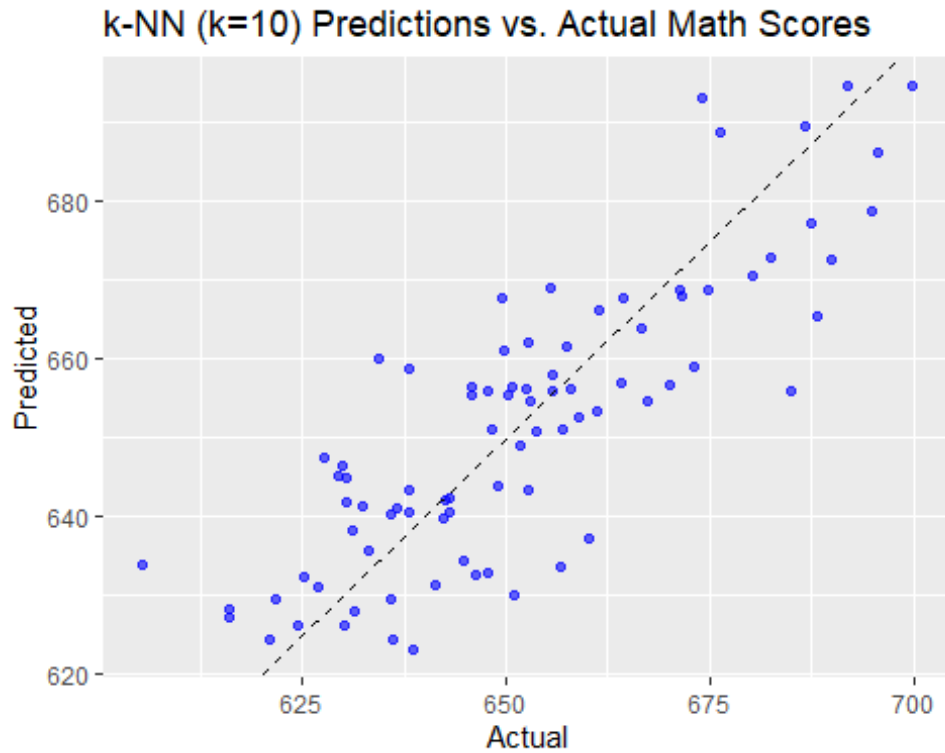
```

Using **k = 10**, the k-NN model performed well, with an R^2 of ~0.69, which is close to the linear model. This suggests k-NN can effectively capture local relationships in the data.

```

#Visualization
ggplot(data.frame(Actual = test_labels, Predicted = predictions),
       aes(x = Actual, y = Predicted)) +
  geom_point(alpha = 0.6, color = "blue") +
  geom_abline(slope = 1, linetype = "dashed") +
  labs(title = "k-NN (k=10) Predictions vs. Actual Math Scores")

```



The scatter plot shows predicted vs. actual math scores. Most points cluster near the diagonal line, which indicates accurate predictions. However, some variation remains, which is expected in real-world data.

Summary:

The k-NN algorithm offers a competitive alternative to linear models for predicting math scores. With proper scaling and choice of k, it yields solid predictive accuracy. For the CASchools dataset:

- $k = 10$ gave the best performance (lowest RMSE).
- While less interpretable, k-NN proves effective in capturing patterns in the data without model assumptions.

Chapter 7: Decision Trees and Regression Trees for Predicting Student Performance.

In this chapter, we employ regression trees to model and predict student performance in both Math and English using school-level factors from the CASchools dataset. Regression trees are powerful tools for capturing non-linear relationships and offering clear, interpretable decision rules that reveal how specific conditions influence academic outcomes.

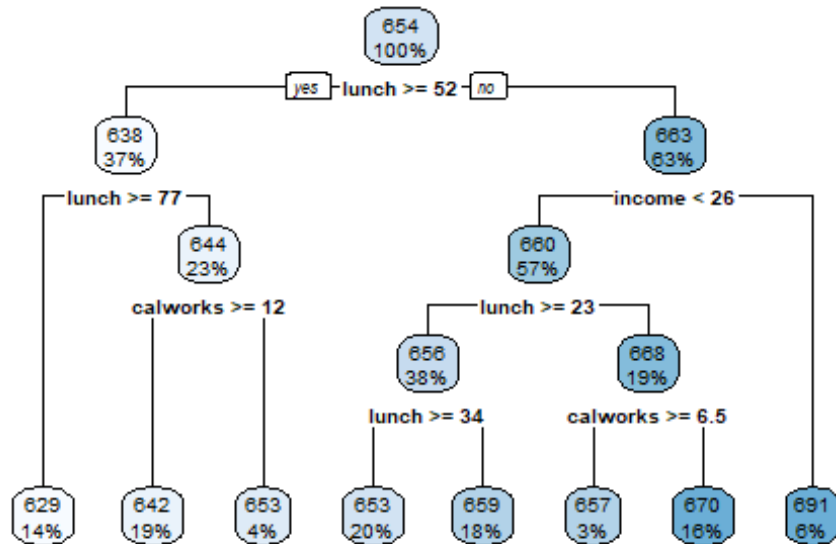
Regression Tree for Math Scores

We first fit a regression tree model to predict math scores based on key school-level variables.

```
# Regression Tree to predict math scores
tree_model <- rpart(math ~ income + english + lunch + calworks + computer + t
eachers+ computer+expenditure,
                    data = train, method = "anova")

# Visualize the tree
rpart.plot(tree_model, main = "Regression Tree for Math Scores")
```

Regression Tree for Math Scores



This tree shows how student math performance is influenced by variables like income, lunch program participation, and computer availability. Each branch represents a rule, and the leaves predict average math scores under those conditions. The tree is easy to interpret and highlights key splits, such as income thresholds or English learner percentages.

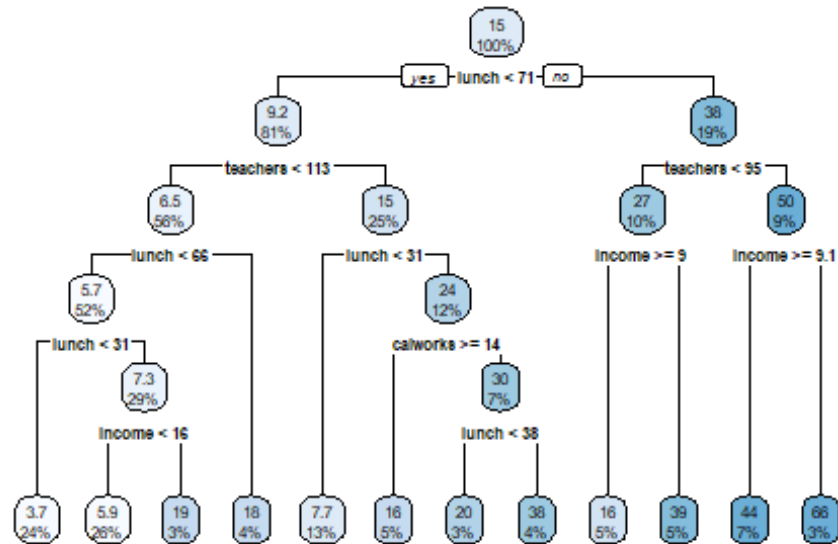
Regression Tree for English Scores

Next, we fit a regression tree to predict English scores, incorporating Math scores as an additional predictor:

```
tree_model2 <- rpart(english ~ income + math + lunch + calworks + computer +
  teachers+ computer+expenditure,
  data = train, method = "anova")

# Visualize the tree
rpart.plot(tree_model2, main = "Regression Tree for English Scores")
```


Regression Tree for English Scores



For English scores, the tree structure again points to socio-economic indicators like income and lunch program status as significant factors. Interestingly, math scores also play a role here, showing how subjects can be interrelated in performance prediction.

Model Evaluation

We assess model performance using the test dataset and compute standard regression metrics.

```

#predictions math score
# Predict on test data
tree_predictions <- predict(tree_model, newdata = test)

# Evaluate performance
postResample(pred = tree_predictions, obs = test$math)

##          RMSE   Rsquared         MAE
## 11.9738510  0.6665559  9.5868752

#predictions english score
# Predict on test data
tree_predictions2 <- predict(tree_model2, newdata = test)

# Evaluate performance
postResample(pred = tree_predictions2, obs = test$english)
  
```

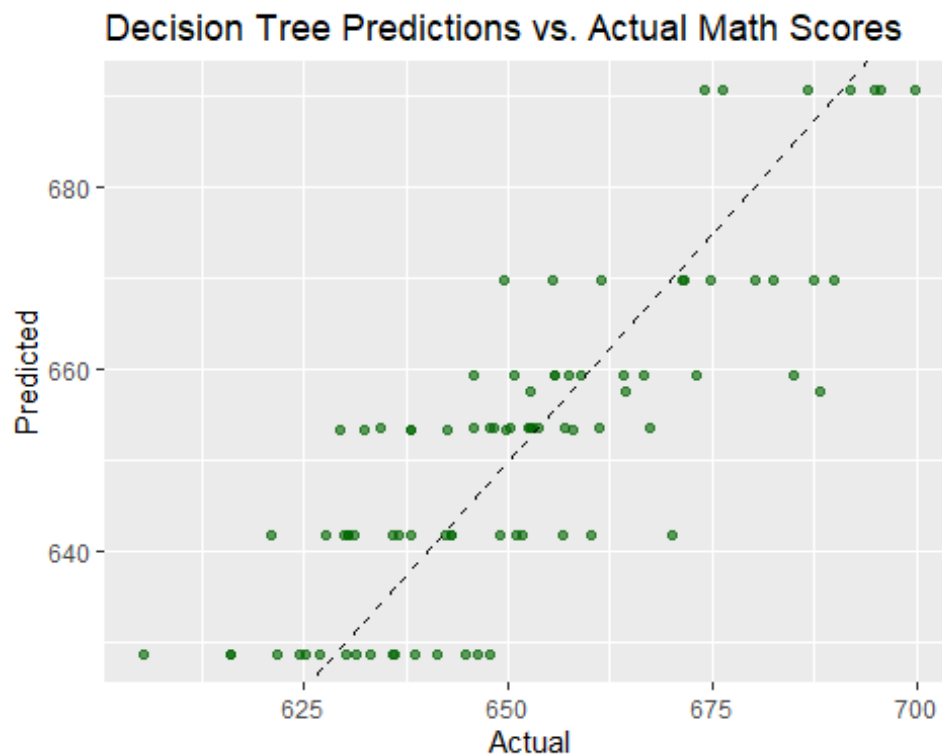
```
##          RMSE    Rsquared      MAE
## 16.1036797  0.4094684 10.6331346
```

The regression tree predicted math scores with a good R^2 of 0.67, meaning it explained 67% of the variance — quite effective for a tree-based model. The English score model had a lower R^2 (0.41), suggesting more variability in English performance that the model couldn't capture as well.

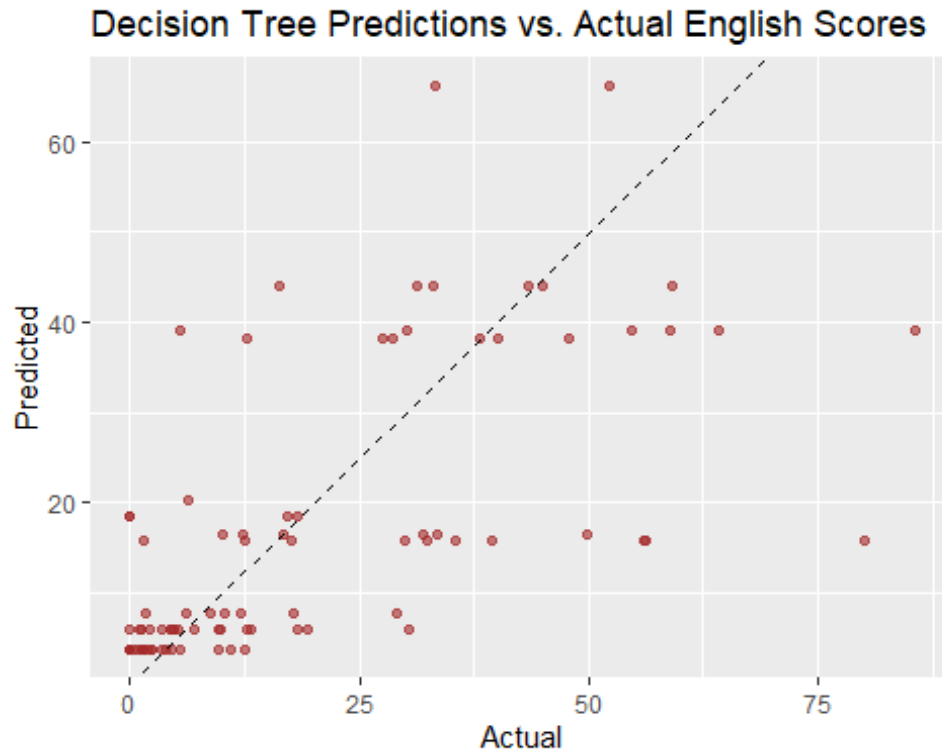
Visualization:

We visualize the agreement between predicted and actual values using scatter plots:

```
#plotting actual vs pred
#math score
ggplot(data.frame(Actual = test$math, Predicted = tree_predictions),
       aes(x = Actual, y = Predicted)) +
  geom_point(alpha = 0.6, color = "darkgreen") +
  geom_abline(slope = 1, linetype = "dashed") +
  labs(title = "Decision Tree Predictions vs. Actual Math Scores")
```



```
#english scr
ggplot(data.frame(Actual = test$english, Predicted = tree_predictions2),
       aes(x = Actual, y = Predicted)) +
  geom_point(alpha = 0.6, color = "brown") +
  geom_abline(slope = 1, linetype = "dashed") +
  labs(title = "Decision Tree Predictions vs. Actual English Scores")
```



- **Math Scores:**

Points mostly follow the diagonal line, indicating strong agreement between predicted and actual scores.

- **English Scores:**

There is more spread around the line, echoing the earlier finding of lower predictive strength for English scores.

Summary:

Regression trees provided an intuitive and interpretable modelling approach for predicting student performance from school-level factors. The math model was notably effective, identifying income, computer availability, and lunch program participation as major predictors. The English model, though insightful, was less predictive, suggesting the need for more language-specific features or more complex modelling techniques.

In summary, regression trees confirm that socio-economic indicators remain strong predictors of academic outcomes, and performance in one subject (e.g., Math) can be informative for others.

Chapter 8: Cluster Analysis

Cluster analysis is a powerful unsupervised learning technique used to group observations with similar characteristics. In this chapter, we apply both K-means and Hierarchical clustering to California school data, using standardized socio-economic and performance-related variables to uncover natural groupings among schools.

Data Preparation

We select key socio-economic variables for clustering and standardize them to ensure equal weighting in distance calculations:

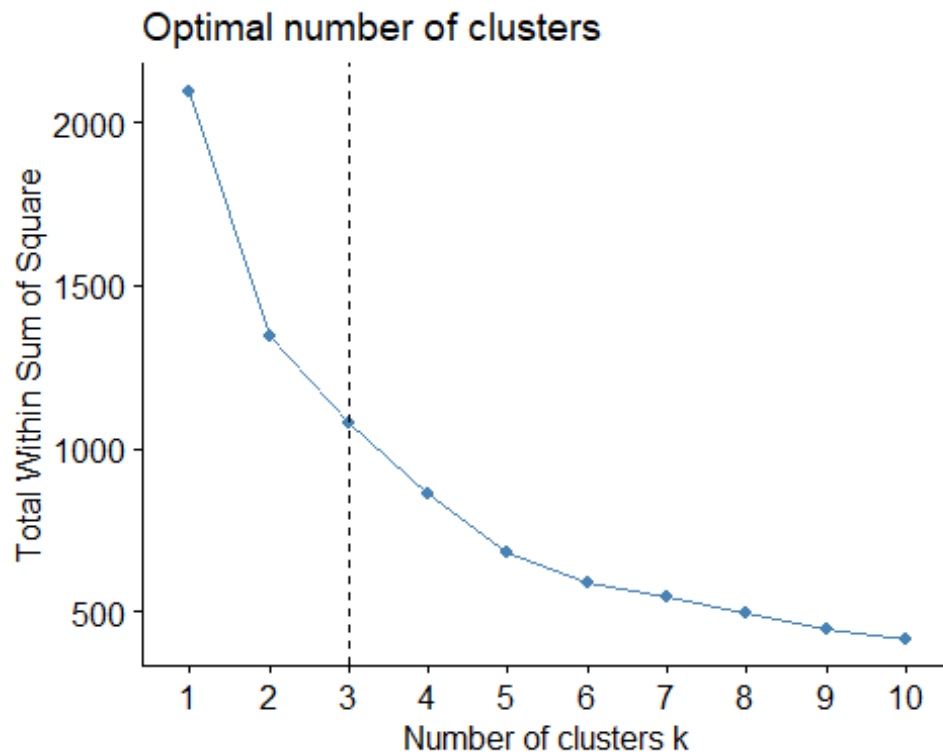
```
features <- c("income", "english", "lunch", "calworks", "computer")
scaled_data <- scale(caschools[, features]) # Standardize for clustering
```

Standardization (mean = 0, standard deviation = 1) is essential when using distance-based clustering methods like K-means or hierarchical clustering to prevent variables with larger scales from dominating the analysis.

K-Means Clustering

To determine the optimal number of clusters, we apply the Elbow Method:

```
#kmeans clustering
fviz_nbclust(scaled_data, kmeans, method = "wss") +
  geom_vline(xintercept = 3, linetype = 2) # Elbow at k=3
```



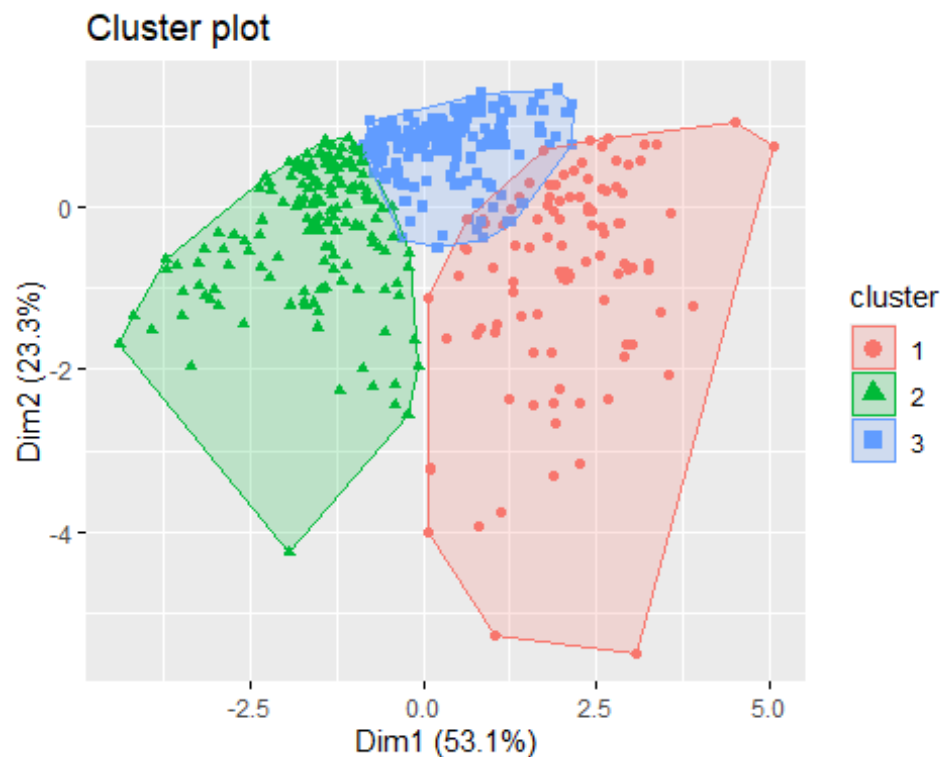
Elbow Method Result:

The plot of Within-Cluster Sum of Squares (WSS) shows a clear elbow at $k = 3$, suggesting three distinct clusters.

We then perform K-means clustering:

```
set.seed(123)
kmeans_result <- kmeans(scaled_data, centers = 3, nstart = 25)

#Visualization
fviz_cluster(kmeans_result, data = scaled_data, geom = "point")
```



Cluster Interpretation(K-Means)

Three distinct groups of schools emerge. The clusters may reflect socio-economic differences, such as:

- **Cluster 1:** Likely high-income, low English learner schools.
- **Cluster 2:** Moderate mix.
- **Cluster 3:** Schools with higher needs (e.g., more lunch subsidies and English learners).

Hierarchical Clustering

Hierarchical clustering helps understand nested structures among schools. We use Ward's method with Euclidean distance, which tends to produce compact, similar-sized clusters.

```
#hierarchical Clustering
dist_matrix <- dist(scaled_data, method = "euclidean")
hc <- hclust(dist_matrix, method = "ward.D2")
```

We used **Ward's method** with **Euclidean distance**, which tends to form clusters of similar size and minimizes variance within clusters.

Dendrogram

To build a more interpretable tree, we aggregate the data into 20 mini-clusters, compute their means, and relabel them meaningfully.

```
# Cut tree into 3 clusters
hc_clusters <- cutree(hc, k = 3)

#dendrogram

# Step 1: Aggregate data by similar schools first
agg_data <- aggregate(caschools[, c("income", "english", "math", "read")],
                      by = list(cluster = cutree(hc, k = 20)), # Pre-group
                      into 20 clusters
                      FUN = mean)

# Step 2: Create dendrogram with enhanced Labels
hc_agg <- hclust(dist(scale(agg_data[, -1])), method = "ward.D2")

# Create meaningful Labels showing key stats
agg_data$label <- paste(
  "Cluster ", agg_data$cluster, "\n",
  "Income: $", round(agg_data$income, 1), "k\n",
  "English: ", round(agg_data$english, 1), "%\n",
  "Math: ", round(agg_data$math, 0),
  sep = ""
)

dend_agg <- as.dendrogram(hc_agg) %>%
  set("branches_k_color", k = 3) %>%
  set("labels", agg_data$label) %>%
  set("labels_cex", 0.7) %>%
  set("labels_col", "darkblue") %>%
  set("branches_lwd", 1.5)

# Step 3: Plot with annotations
par(mar = c(5, 2, 1, 15)) # Increase right margin for Labels
plot(dend_agg, horiz = TRUE,
     main = "Hierarchical Clustering of California Schools\n(Grouped by
Income, English Learners, and Test Scores)",
     xlab = "Dissimilarity (Ward's Method)")

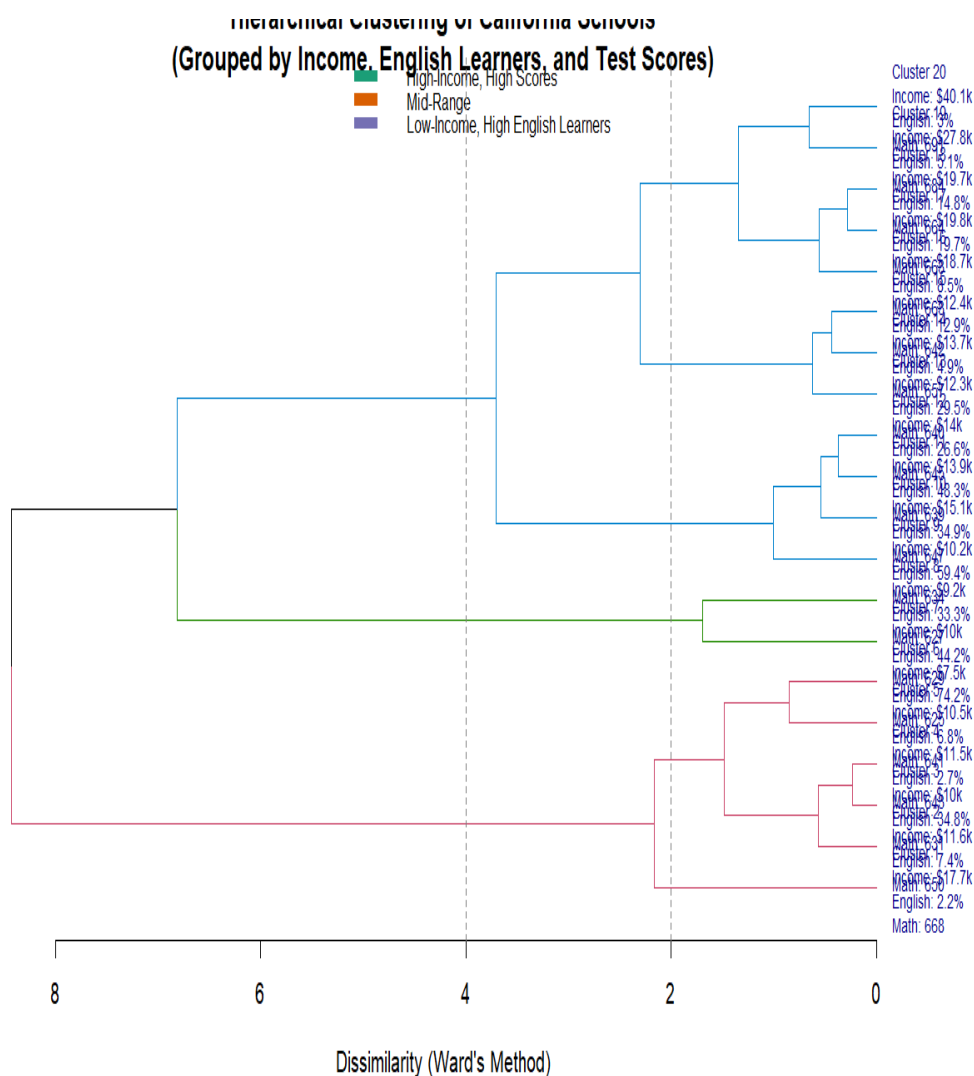
# Add cluster interpretation
```

```

legend("topright",
      legend = c("High-Income, High Scores",
                  "Mid-Range",
                  "Low-Income, High English Learners"),
      fill = c("#1b9e77", "#d95f02", "#7570b3"), # Match dendrogram colors
      border = NA,
      bty = "n",
      cex = 0.8)

# Add reference lines
abline(v = c(2, 4), lty = 2, col = "gray50")

```



Summary:

Both **K-means** and **Hierarchical clustering** revealed three major types of schools:

- **Cluster A:** High-income, high-performance.
- **Cluster B:** Moderate characteristics across features.
- **Cluster C:** Low-income, high English learner percentage, lower test scores.

These clusters are useful for targeting educational policy and funding efforts more effectively.

Chapter 9: Linear Discriminant Analysis(LDA) & Quadratic Discriminant Analysis(QDA)

To gain deeper insights into student performance, we introduce a binary classification problem. We group students into "High" or "Low" math performers based on whether their scores exceed the median. This setup allows us to apply classification techniques like LDA and QDA to predict math performance groups using key school-level features.

Creating the Target Variable

```
# Create binary math performance groups
caschools$math_group <- ifelse(caschools$math > median(caschools$math),
"High", "Low")
caschools$math_group <- as.factor(caschools$math_group)
```

LDA (Linear Discriminant Analysis)

```
#LDA
lda_model <- lda(math_group ~ income + english + expenditure_per_student,
data = train)
lda_pred <- predict(lda_model, test)
confusionMatrix(lda_pred$class, test$math_group)

## Confusion Matrix and Statistics
##
##              Reference
## Prediction High Low
##      High    29   8
##      Low     9   38
##
##              Accuracy : 0.7976
##              95% CI : (0.6959, 0.8775)
##      No Information Rate : 0.5476
##      P-Value [Acc > NIR] : 1.514e-06
##
##              Kappa : 0.5906
##
##      McNemar's Test P-Value : 1
##
```

```
##           Sensitivity : 0.7632
##           Specificity : 0.8261
##           Pos Pred Value : 0.7838
##           Neg Pred Value : 0.8085
##           Prevalence : 0.4524
##           Detection Rate : 0.3452
##           Detection Prevalence : 0.4405
##           Balanced Accuracy : 0.7946
##
##           'Positive' Class : High
##
```

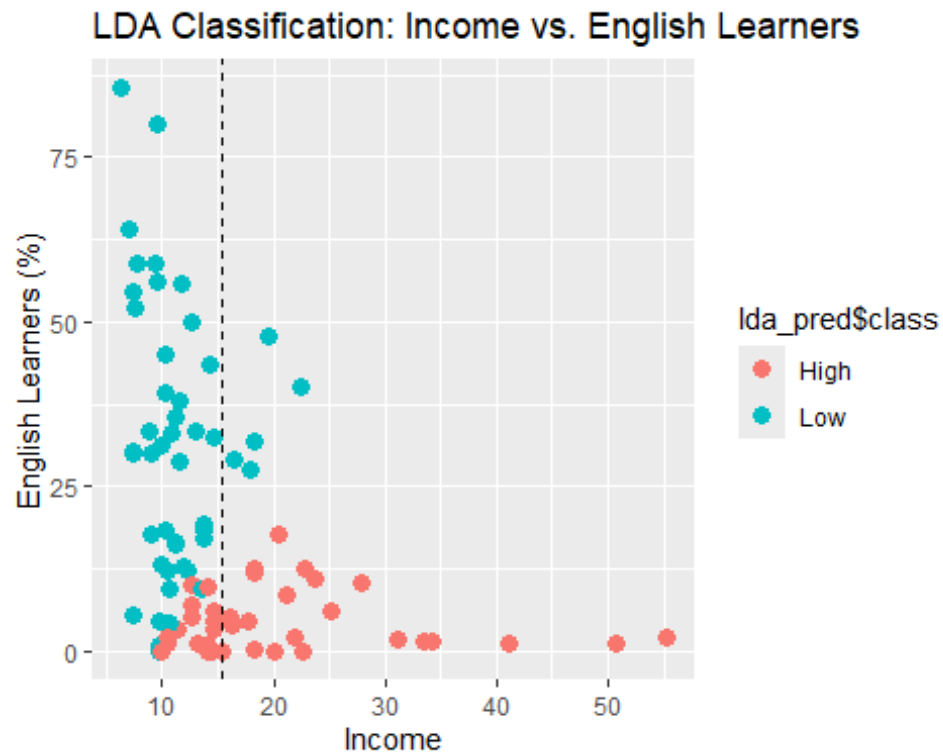
Key Performance Metrics:

- **Accuracy:** 79.8%
- **Balanced Accuracy:** 79.5%
- **Sensitivity (Recall for "High"):** 76.3%
- **Specificity:** 82.6%
- **Kappa:** 0.5906

These results show that LDA performs well at separating high and low math performers using the chosen features, with fairly balanced sensitivity and specificity.

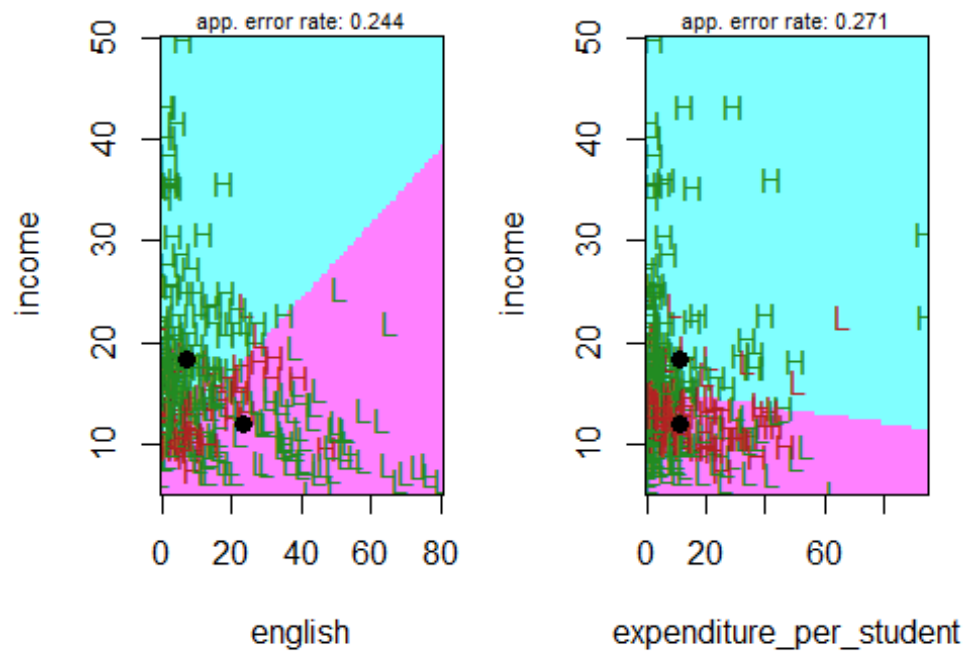
#plot LDA

```
ggplot(test, aes(x = income, y = english, color = lda_pred$class)) +
  geom_point(size = 3) +
  geom_vline(xintercept = mean(caschools$income), linetype = "dashed") +
  labs(title = "LDA Classification: Income vs. English Learners",
       x = "Income", y = "English Learners (%)")
```

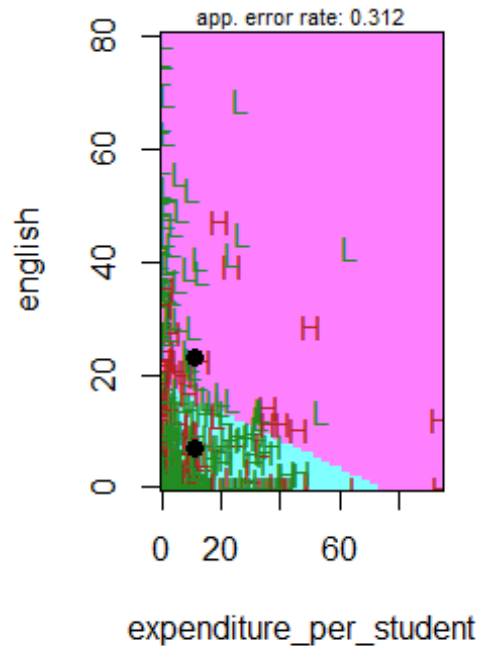


This scatter plot illustrates how LDA separates groups based on income and English learner percentages. The dashed line shows the average income, giving visual insight into class distribution.

```
For LDA
partimat(math_group ~ income + english + expenditure_per_student,
  data = train,
  method = "lda",
  main = "LDA Classification Boundaries",
  col.correct = "forestgreen", # Correct classifications
  col.wrong = "firebrick")    # Misclassifications
```



LDA Classification Boundaries



The `partimat()` plot shows clear decision boundaries — green areas mark correct predictions, while red spots highlight misclassified instances.

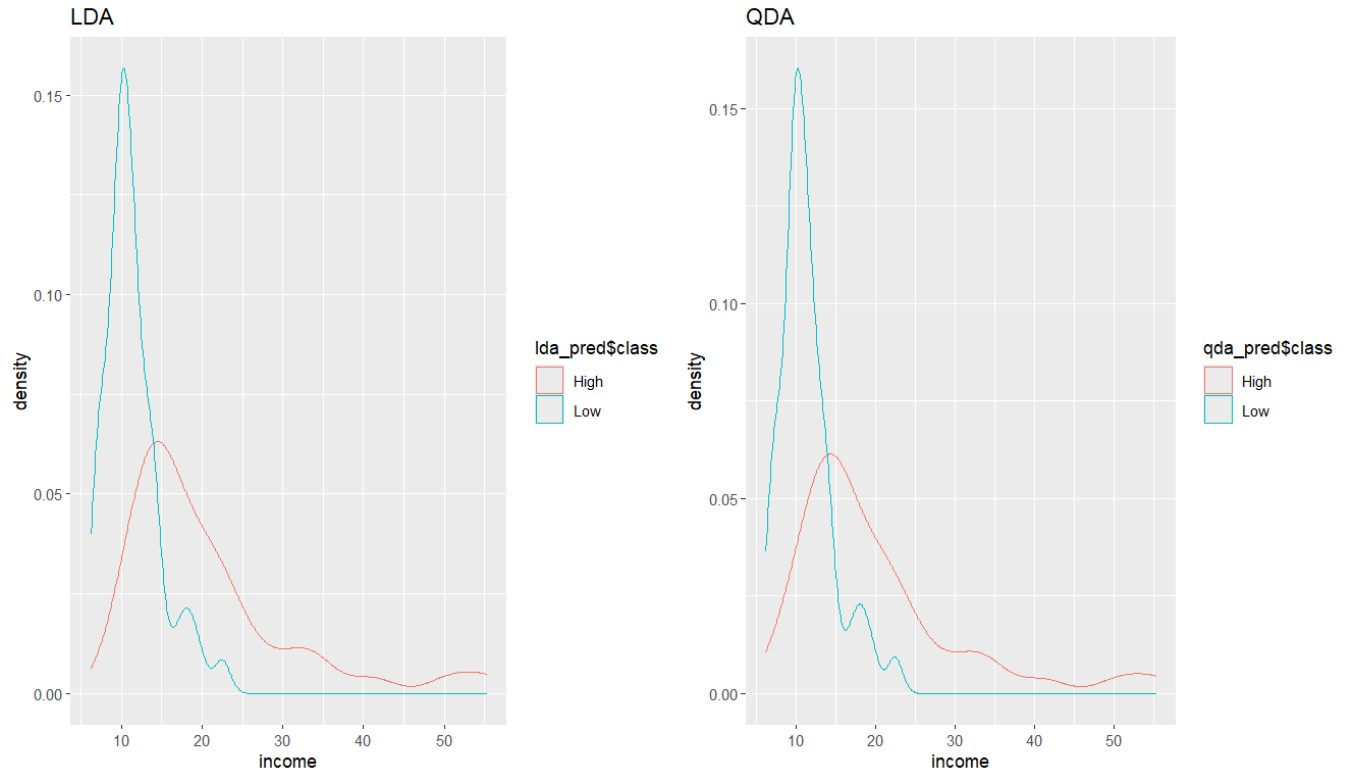
QDA (Quadratic Discriminant Analysis)

```
#QDA
qda_model <- qda(math_group ~ income + english + expenditure_per_student, data = train)
qda_pred <- predict(qda_model, test)
confusionMatrix(qda_pred$class, test$math_group)

## Confusion Matrix and Statistics
##
##              Reference
## Prediction High Low
##           High   31   8
##           Low    7  38
##
##              Accuracy : 0.8214
##              95% CI : (0.7226, 0.8965)
##      No Information Rate : 0.5476
##      P-Value [Acc > NIR] : 1.215e-07
##
##              Kappa : 0.6404
##
##  Mcnemar's Test P-Value : 1
##
##              Sensitivity : 0.8158
##              Specificity : 0.8261
##              Pos Pred Value : 0.7949
##              Neg Pred Value : 0.8444
##              Prevalence : 0.4524
##              Detection Rate : 0.3690
##      Detection Prevalence : 0.4643
##              Balanced Accuracy : 0.8209
##
##              'Positive' Class : High
##
```

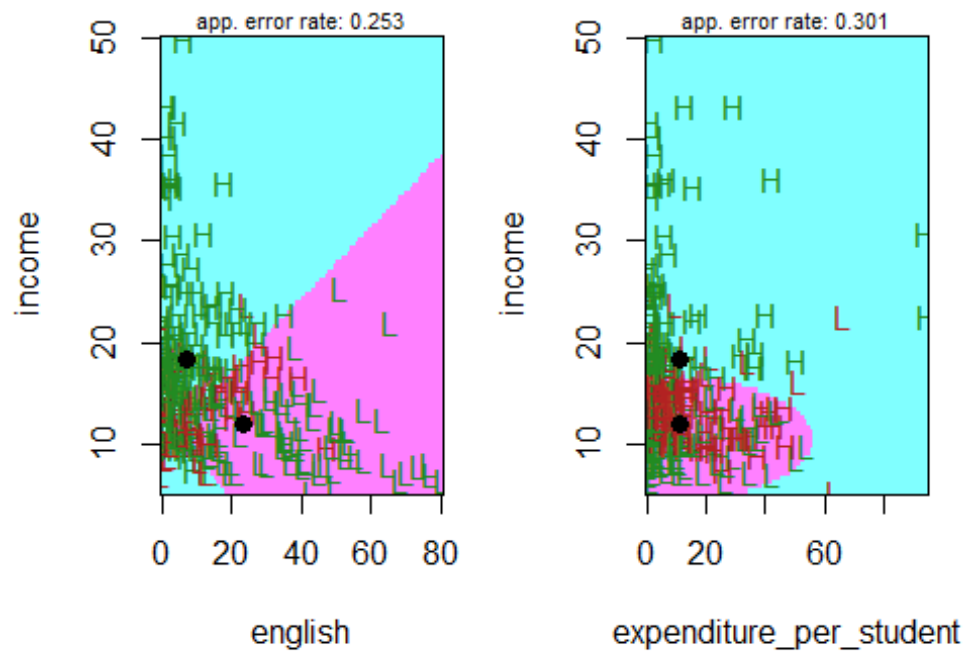
QDA slightly outperforms LDA, particularly in sensitivity and overall accuracy, indicating better handling of more complex, non-linear relationships.

```
plot1 <- ggplot(test, aes(x = income, color = lda_pred$class)) + geom_density() + ggtitle("LDA")
plot2 <- ggplot(test, aes(x = income, color = qda_pred$class)) + geom_density() + ggtitle("QDA")
grid.arrange(plot1, plot2, ncol = 2)
```

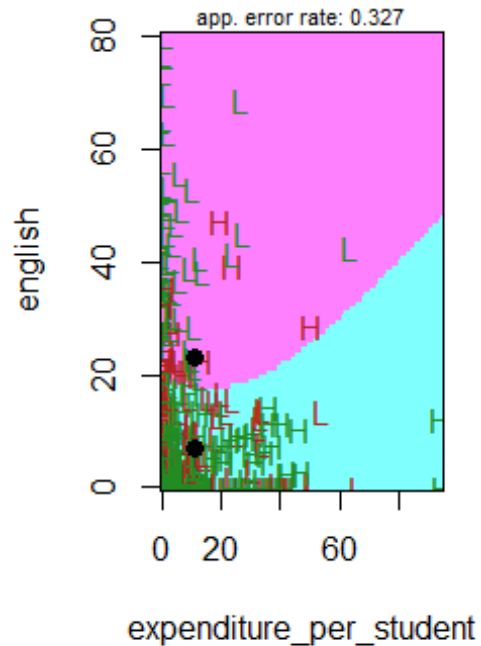


These density plots show how predictions vary by model. QDA shows slightly sharper class separation for income, highlighting its flexibility in fitting curved boundaries.

```
partimat(math_group ~ income + english + expenditure_per_student,
  data = train,
  method = "qda",
  main = "QDA Classification Boundaries",
  col.correct = "forestgreen",
  col.wrong = "firebrick")
```



QDA Classification Boundaries



The QDA partition plot shows finer classification boundaries and generally fewer misclassifications compared to LDA.

Summary:

Model	Accuracy	Sensitivity	Specificity	Balanced Accuracy	Kappa
LDA	79.8%	76.3%	82.6%	79.5%	0.5906
QDA	82.1%	81.6%	82.6%	82.1%	0.6404

Both LDA and QDA are effective tools for predicting student math performance using school-level data. While LDA performs well with linear relationships, QDA's flexibility allows it to capture subtler, curved boundaries — resulting in slightly better performance.

These classification models can support educational interventions, helping identify schools or student groups that may benefit from additional support based on income levels, English learner rates, and expenditure patterns.
